

Ming-Shan Lee

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[LinkedIn](#) | [Portfolio](#)

EDUCATION

University of Southern California (USC)

Aug 2023 – May 2025

Master of Science in Applied Data Science

Los Angeles, CA

Relevant Coursework: Data Management, Data Science at Scale, Data Mining, Deep Learning

National Tsing Hua University (NTHU)

Sept 2019 – June 2023

Bachelor of Science in Industrial Engineering and Engineering Management

Hsinchu, Taiwan

Relevant Coursework: Discrete Mathematics, Engineering Statistics, Linear Algebra, Data Structures

WORK EXPERIENCE

University of Southern California

Sept 2025– Present

Volunteer Research Assistant

- Collaborating on research related to **Model Context Protocol (MCP)** with a focus on open-source community contributions.

Syncratik

Sept 2025– Present

Full Stack Web Developer Intern

- Developing full-stack dashboards (**Node.js**, **REST APIs**, **Firebase**) for clients in a startup environment.

Micron Technology

July – Aug 2024

Data Scientist Intern

- Developed and optimized defect prediction machine learning model for semiconductor manufacturing using **Google BigQuery**, achieving **20% improvement in R-squared value** through data mining and factor interaction analysis, with optimized model deployed to production.
- **Implemented advanced evaluation metrics and data visualization frameworks** using **Bitbucket** for version control, enabling seamless collaboration and delivering actionable insights to manufacturing stakeholders.

NTHU Lean Management Laboratory

May – Sept 2022

Research Assistant

- Led **machine learning implementation** for semiconductor company's inaugural ML project, strategically securing comprehensive datasets and developing a hybrid system with custom algorithms alongside SVR, Random Forest, and XGBoost models.
- **Achieved 25% superior performance** compared to existing implementations across evaluated product lines through pioneering algorithm for imbalanced data challenges in Industry-Academia partnership.

SKILLS

Programming Languages: C, C++, Python (NumPy, Pandas, Scikit-learn, PySpark), Java, SQL, HTML, CSS, JavaScript

Data Management & Visualization: Tableau, MySQL, NoSQL, MongoDB, BigQuery

Cloud & Big Data Technologies: Google Cloud Platform (GCP), AWS, Hadoop

CERTIFICATIONS

Salesforce Certified Tableau Desktop Foundations | July 2025

PROJECTS

Better Agent for AppWorld | Python, LLM, Reinforcement Learning, Fine-Tuning

Aug – Dec 2024

- A project that aims to improve LLM model's performance in AppWorld, a platform that provides LLM agents an environment to test their ability in solving day-to-day tasks.
- Enhanced LLaMA-3.1 8B Instruct on AppWorld using SFT and RLHF (PPO, DPO), achieving 6.6% Task Goal Completion (5.4× over baseline) with SFT outperforming RLHF and hybrid methods.
- GitHub: github.com/Aragath/Better-Agent-For-AppWorld

Hybrid Yelp Rating Predictor | Python, Spark(PySpark), Recommendation System, NLP

January – May 2025

- Built a hybrid recommendation system to predict Yelp user ratings for unseen restaurants, services, and businesses.
- Combined collaborative filtering, content-based features, sentiment analysis, and XGBoost, reducing test RMSE from 1.2 to 0.9 and effectively handling cold-start challenges.
- GitHub: github.com/Atatsuki2000/Hybrid-Yelp-Rating-Predictor